

RESOURCE AND ENVIRONMENT INSTRUMENT LEADER

Geopro T-Series 2D Ground Penetrating Radar

Introductory manual

CONTENTS

Company Introduction

Chapter 1

Chapter 2

Products & Solutions

Case Studies

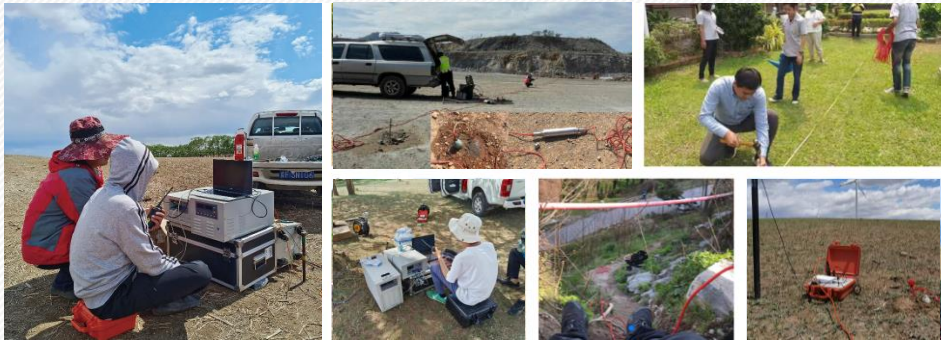
Chapter 3

RESOURCE AND ENVIRONMENT INSTRUMENT LEADER

Geotech Instrument Co., Ltd. is a high-tech enterprise integrating geophysical exploration and engineering geophysical equipment sales and technical services, and is a comprehensive service provider in the field of geophysical exploration and engineering geophysical exploration.

Geotech is committed to open cooperation with partners to build more intelligent equipment solutions in the field of geophysical exploration and engineering geophysical exploration. Geotech's product system covers **electrical**, **magnetic**, **seismic**, **radar** and other series, including **electrical exploration instruments**, **magnetic exploration instruments**, **seismographs**, **geological radars**, **well logging exploration instruments** and other geological exploration instruments.

Geotech has established good cooperative relations with many domestic and foreign units, and has provided relevant products and services to many engineering quality inspection units, survey units, design units, and scientific research units, and has established long-term and stable cooperative relations.



RESOURCE AND ENVIRONMENT INSTRUMENT LEADER



Company Honors



System certification



Product certification



Electrical
Instrument



MT



SSIP



Geological Radar



Seismic Instrument



Magnetic
Instrument



Well Logging



Borehole Camera

CONTENTS

Company Introduction

Chapter 1

Chapter 2

Products & Solutions

Case Studies

Chapter 3

2D Ground Penetrating Radar

Geopro T-Series





01 Abstract

Geopro T-Series is one portable, intelligent, high performance and full digitalize ground penetrating radar main machine, compatible Geopro T whole series antennas, support most of the GPS equipment, stable and reliable complete Chinese collect system which based on Windows system are in accordance with operating habits of field operators, it has advantages at easy to carry, simple operation, better data quality and high collect efficiency, etc, widely applied in tunnel engineering, road engineering, water conservancy engineering, bridge engineering, civil engineering, military and archaeology areas.

02 Features

- Intelligent host, automatically identifies antenna type and configures radar parameters.
- Saves original data and has the function of restoring on-site data.
- Supports external GPS to add geographic coordinate information to radar data.
- Digital signal transmission is used between the host and the antenna, with strong anti-interference ability and high data signal-to-noise ratio.



03 Application areas



Tunnel engineering

Lining, laying quality inspection, outstrip early report, inverted arch quality inspection, etc.



Road engineering

Structure layer thickness inspection, cracks inspection and roadbed fail and harm inspection, etc.



Water conservancy engineering

Water source research, ice layer detection and water bottom structure detection, etc



Bridge engineering

Concrete iron fix position, steel twist wire fix position, concrete fail and harm inspection, etc



Civil engineering

Pipeline fix position, road cavity inspection and road structure inspection, etc



Geology survey

Rock melt detection, frozen earth detection and stratum form detection, etc



Military

UXO detection, underground defense works detection and life detection, etc



Archaeology

Historic ruins detection imaging and guide digging etc

04 Technical Parameters

Host

Radar index	
Item	Parameters
Antenna	Compatible Geopro T whole series antennas
Channel quantity	1 channel
Memory space	128G
Indicator	10.4", 1024×768 pixel resolution ratio, 32 bytes color LED display screen
GPS	(Series RS232), support GPGLL data format
Display mode	False color, gray level, stacking wave and color stacking wave
Power supply	Interior install lithium battery(NI2040)
Power consume	20W
Continue working time	5 hours(actual working time relate to working temperature, antenna types and cable length)

Data collecting	
Data format	SRD
Modulus adapter bytes	16 bit
The max scanning speed	512 scanning/s(sampling points number is 512)
Sampling points number	256, 512, 1024, 2048, 4096, 8192
The max launch frequency	800kHz
Collecting mode	Time mode, distance mode and point measure mode
Time window range	1~8000ns
Gain range	-10dB~160dB
Gain adjustment point quantity	9 points
Real time process function	Correction of zero bias, FIR filtering, IIR filtering, background elimination, channel averaging, data superposition

Exterior installed terminal	
Data transmission	USB 3.0
GPS	Series (RS232)
Main control	8 needles aviation socket
Machine index	
Volume	314mm×238mm×64mm
Weight	3kg (without battery)
Environment index	
Working temperature	-20℃~+50℃
Storage temperature	-40℃~+60℃
Working humidity	<95%, not condensate
Protection level	IP65

Antenna

Geopro T-series antenna of Geotech has various types, complete function and wide covered frequency range, center frequency 100MHz~2500MHz, these able to meet user's different requirements under multiply detection scenes, the detail parameters as below:

T2500 ground coupling antenna				
Volume	250mm×170mm×150mm	Weight	1.3kg	
Center frequency	2500MHz	The max launch frequency	400kHz	
Depth range	0.1~0.3m	Coupling method	Ground coupling	
Application areas	Building structure detection, road surface layer detection and bridge engineering, etc.			
T1600 ground coupling antenna				
Volume	250mm×170mm×150mm	Weight	1.3kg	
Center frequency	1600MHz	The max launch frequency	400kHz	
Depth range	0.2~0.5m	Coupling method	Ground coupling	
Application areas	Building structure detection, road surface layer detection and bridge engineering, etc.			
T900 ground coupling antenna				
Volume	316mm×216mm×170mm	Weight	2.3kg	
Center frequency	900MHz	The max launch frequency	400kHz	
Depth range	0.5~1.5m	Coupling method	Ground coupling	
Application areas	Tunnel lining, laying quality inspection, roadbed fail and harm inspection, building structure detection, etc			

T400 ground coupling antenna				
Volume	316mm×316mm×230mm	Weight	3.6kg	
Center frequency	400MHz	The max launch frequency	400kHz	
Depth range	1~4m	Coupling method	Ground coupling	
Application areas	Tunnel lining, laying quality inspection, roadbed fail and harm inspection, city pipeline detection, water source and hydroelectricity engineering, etc.			
T200 ground coupling antenna				
Volume	500mm×500mm×230mm	Weight	6.4kg	
Center frequency	200MHz	The max launch frequency	400kHz	
Depth range	1~4m	Coupling method	Ground coupling	
Application areas	Tunnel lining, laying quality inspection, inverted arch quality inspection, city pipeline detection, water source and hydroelectricity engineering, roadbed fail and harm inspection, building structure detection, ground survey etc.			
T100 ground coupling antenna				
Volume	980mm×560mm×320mm	Weight	13.6kg	
Center frequency	100MHz	The max launch frequency	100kHz	
Depth range	1~30m	Coupling method	Ground coupling	
Application areas	Tunnel geology outstrip early report, mineral survey, water source and hydroelectricity engineering, etc.			

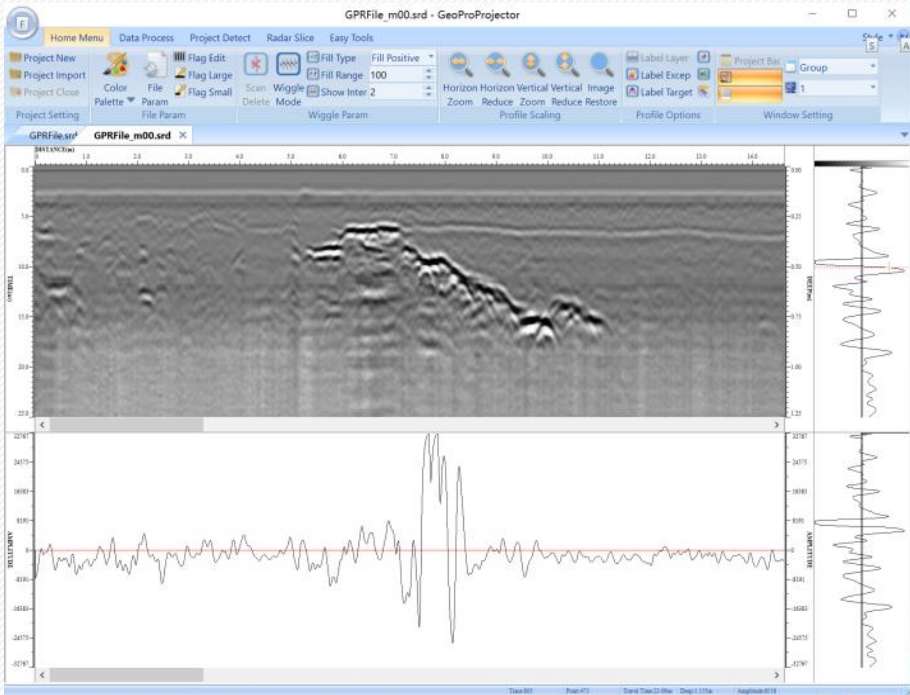
05 Ground penetrating radar data process explain software

Geopro T-series ground penetrating radar data process explain software Geopro Projectoris one syn data process software which aim at Geopro T-series common ground penetrating radar, able to process the operations like data transmit, data loading, data display and data process the ground penetrating radar data which collected by Geopro T-series products and international main other formats ground penetrating radar data, convenient for user find the abnormal information in the radar data and generate the follow report documents which are in accordance with national standards, this is one important tool for Geopro T-series ground penetrating radar data process and analyse. The whole interface of software shown as the picture:

Functions

The software divided into “Menu column”, “Engineering column and source manager column”, “Radar section area”, “Amplitude curve, layer thickness curve, abnormal distribute diagram output column”, “Single way column” and “Status column”, total 6 display areas, finish the syn analyse the radar data through check the content in each area. The main function of this software as below:

Document management	Open, close and save, data split and merge, data format transmit, bitmap output, batch process.
Homepage menu	Engineering setting, document parameter, stacking wave shape, section rooming, section edit, window method.
Data process	Section process, data standard, zero pint gain, data filtering wave, high grade process.
Engineering detection	Layer position detection, abnormal detection, target detection.
Section imaging	Assembly data, data process.
Convenient tools	Help file, edition instruction, common tools.



Features

● **Project-Based Data Orchestration**

Aggregate multi-phase survey data per project, auto-generate time-series comparisons for road subsidence/pipeline aging analysis

80% faster data retrieval vs. traditional folder systems

● **3D Visualization Powerhouse**

Pseudocolor mapping: Precisely outline bedrock/void boundaries (e.g., 2m-diameter void at 3m depth)

Color-stacked wave algorithm: Enhances low-frequency antenna data clarity, resolves sand-clay layers at 50m depth

40% higher imaging efficiency than industry average

● **No-Code Intelligent Processing**

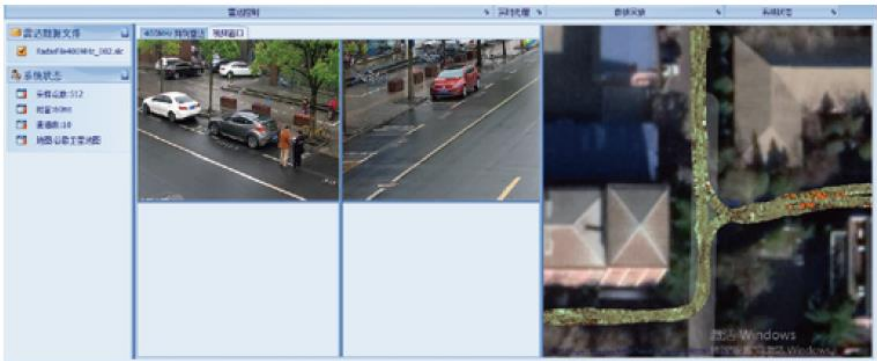
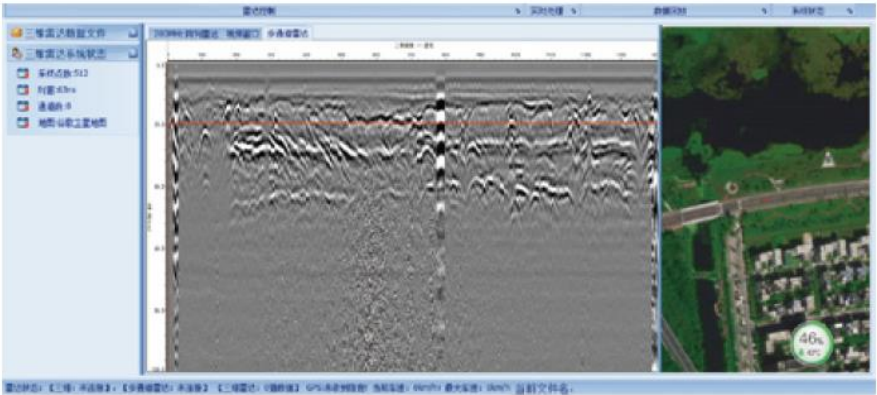
20+ built-in international algorithms (ASTM/GB50021), dual-mode operation (3-step simple mode & professional mode)

Batch processing: Auto-analyze 100+ GPR datasets in one click, 5x faster night-shift efficiency

● **ISO-Standard Report Generator**

Auto-detect layer thickness (error $\leq 0.5\text{cm}$) and defect zones, supports manual refinement

Built-in [Urban Road Code] templates, generate 3D-report in 10min, cut 80% paperwork



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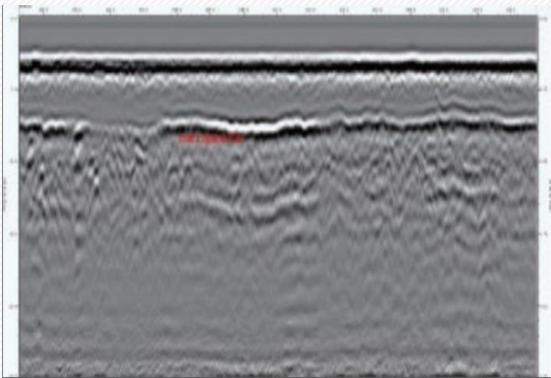
Chapter 3

06 Road application cases

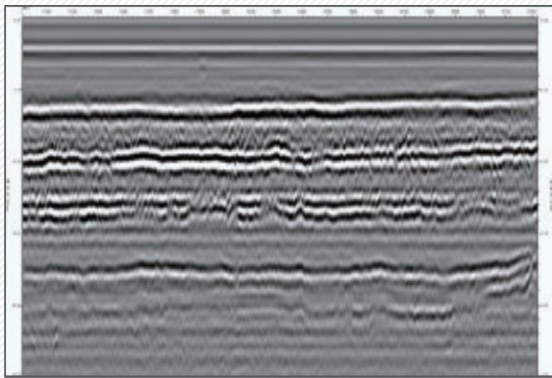
Adopt Geopro T900 antenna process quick detection at distance about 6cm aim at concrete structure road’s high damping characteristic road, able to continue and quickly detect the concrete thickness in the concrete road and defaults, process whole evaluation at concrete road. Also can adopt this method to detect the road structure layerand deep roadbed, obtain the structure layer thickness and defaults of road.



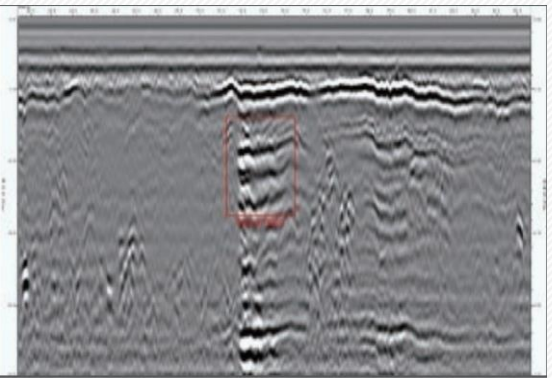
Adopt Geopro T900 ground penetrating radar to process road inspection



Geopro T900 concrete thickness inspection data



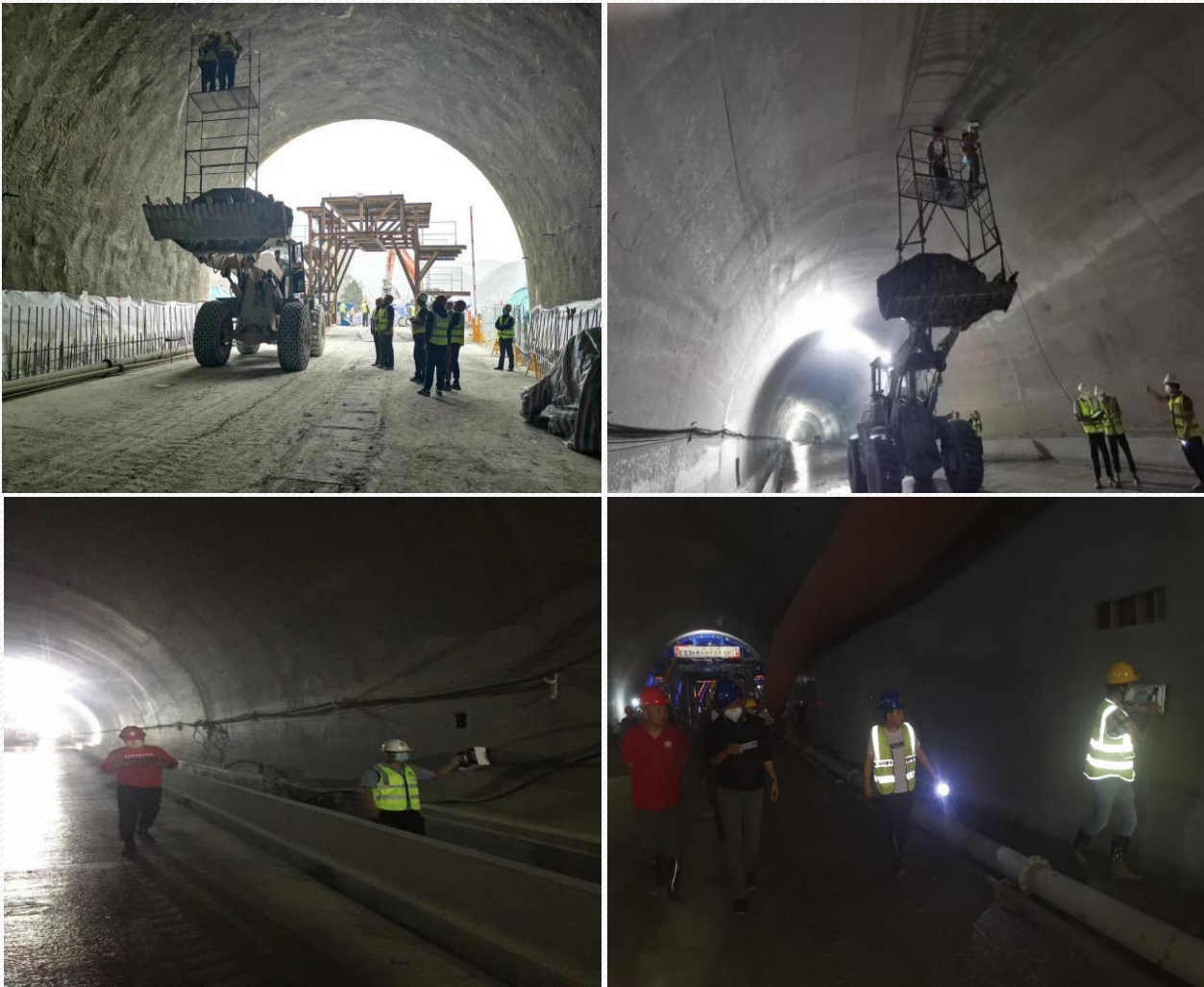
One high speed Geopro T900 antenna road structure inspection



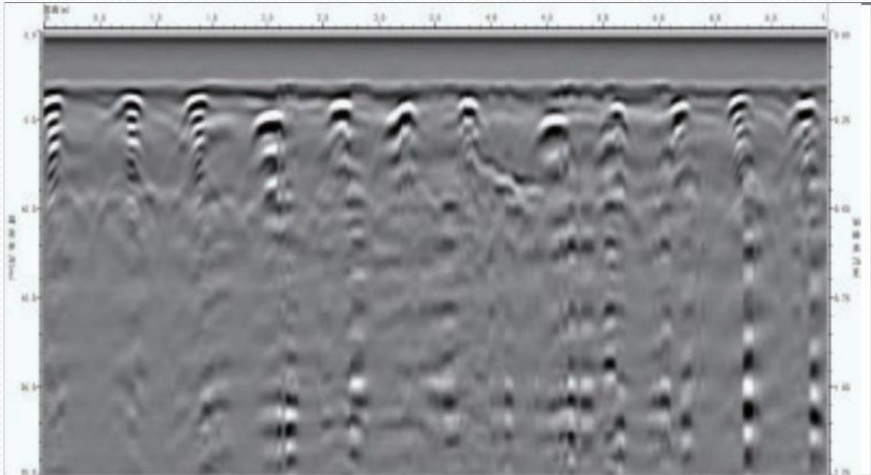
Space under the Geopro T900 antenna concrete road plate

Ground penetrating radar as one main scientific instrument of tunnel engineering inspection, widely applied in the geology outstrip early report, lining, laying quality inspection, inverted arch quality inspection and the lining, laying harm inspection (empty, cavity and not density) during operation. The ground penetrating radar not only can improve the safety factor of tunnel engineering inspection, but also validly guarantee the scientific and reliable of tunnel engineering inspection data, very important performance in the tunnel engineering inspection application.

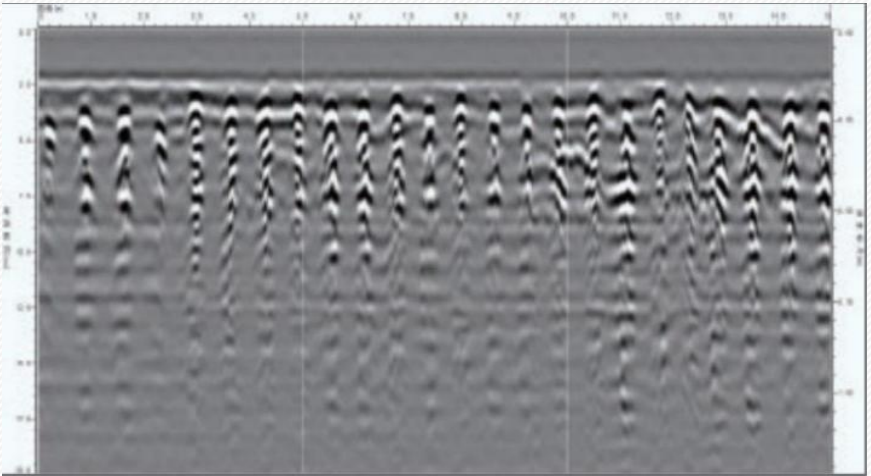
Tunnel field detection



Initial support inspection



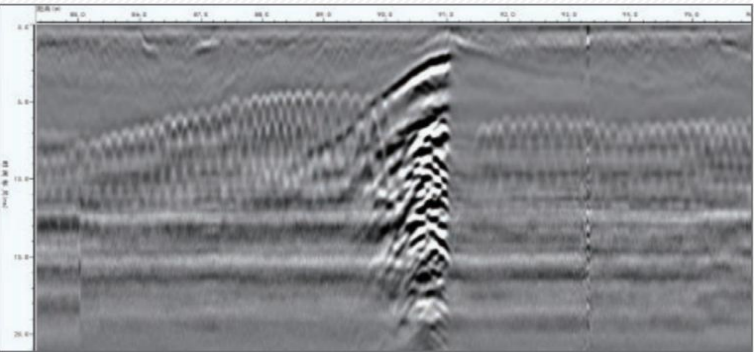
Tunnel initial support arch frame distribution and harms fix position



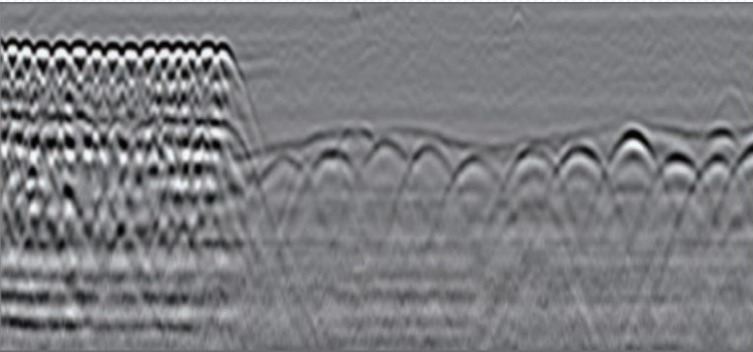
Tunnel initial support arch frame distribution and harms fix position

Second lining inspection

Use 900MHz and 400MHz antenna can inspect the second lining of concrete iron second lining and harms in the concrete iron second lining (empty, cavity and not solid) and lining, laying thickness.



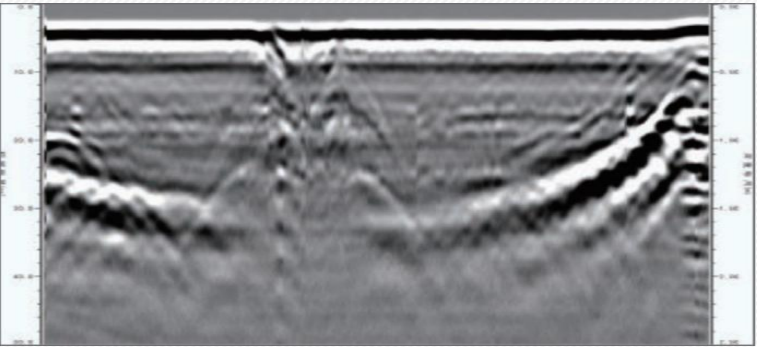
Cavity harms of concrete iron second lining



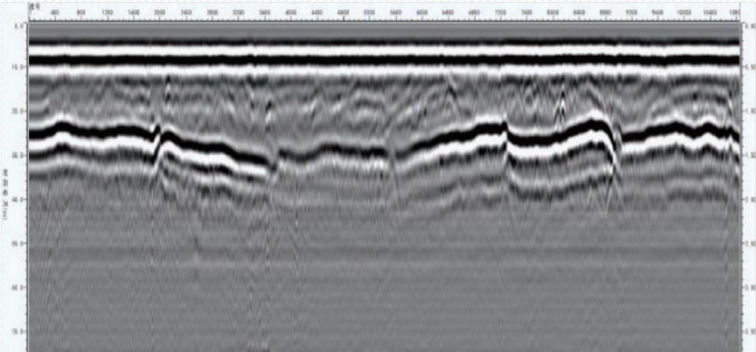
Bottom boundary of second lining and arch frame distribution

Inverted arch

Use Geopro T400 and Geopro T200 antenna can inspect the thickness of inverted arch concrete and concrete not solid.



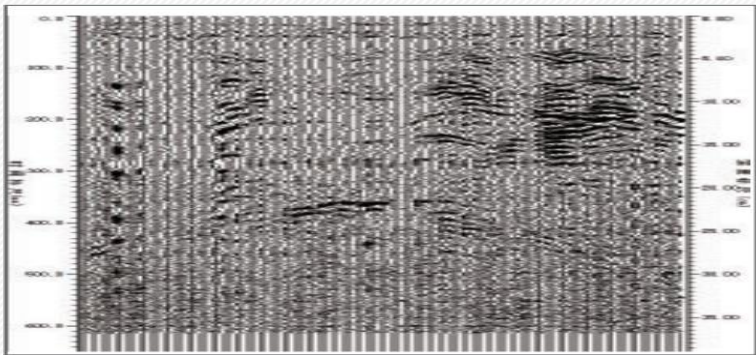
Tunnel bottom inverted arch transverse cut



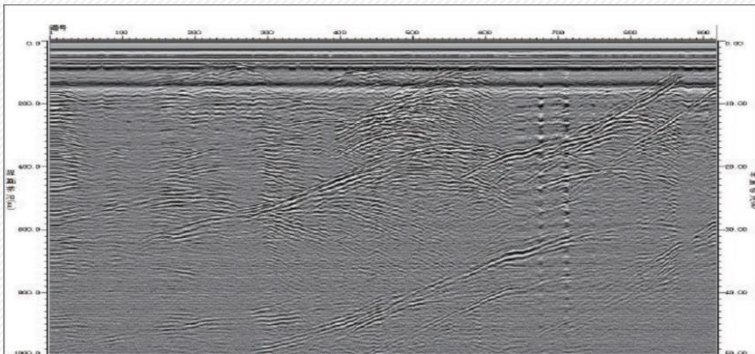
Tunnel bottom inverted arch longitudinal cut radar data

Outstrip early rep

Use 100MHz antenna at point detection and big depth overlap mode can valid detect the geology harms like rock melt, broken layer and crack which at the front of tunnel face.



Outstrip early report-large scale rock

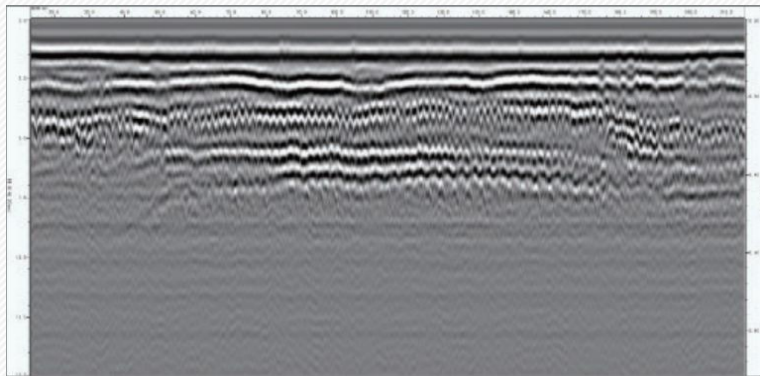


Tunnel bottom melt cavity detectionmelt in limestone

Bridge and building structure



Geopro T1600 ground penetrating radar



Bridge inspection Bridge detection data

Thanks!

Geotech Instrument Co., Ltd.

Mob: +86-15815513204

Email: sales@geotechcn.net

Web: www.geotechcn.net

Whatsapp: +86-15815513204

Wechat: +86-15815513204

Add: Office 403,Tangkeng Community, Longgang District, Shenzhen,China



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